

ASSESSMENT TOOL 1 -- PART 1

Code 1

Markov Decision Process Demonstration

```
grid_size <- 4

start <- c(1,1)
goal <- c(4,4)

state <- start

path <- matrix("",grid_size,grid_size)

path[start[1],start[2]] <- "S"
path[goal[1],goal[2]] <- "G"

print("Initial Grid")
print(path)

step <- 1

cat("\nStarting Simulation\n\n")

while(!(all(state==goal))){

  row <- state[1]
  col <- state[2]

  possible <- list()

  if(row>1)
    possible$Up <- c(row-1,col)

  if(row<grid_size)
    possible$Down <- c(row+1,col)

  if(col>1)
    possible$Left <- c(row,col-1)

  if(col<grid_size)
    possible$Right <- c(row,col+1)

  best_action <- NULL
  best_distance <- 100

  for(a in names(possible)){

    next_state <- possible[[a]]

    distance <- abs(goal[1]-next_state[1]) +
      abs(goal[2]-next_state[2])

    if(distance < best_distance){

      best_distance <- distance
      best_action <- a
      state_next <- next_state

    }

  }

  reward <- -1

  if(all(state_next==goal))
    reward <- 100
```

```

cat("Step:",step,"\n")

cat("Current State :",state,"\n")
cat("Action Taken :",best_action,"\n")
cat("Next State  :",state_next,"\n")
cat("Reward      :",reward,"\n\n")

state <- state_next

step <- step+1

}

```

OUTPUT:

```

[1] "Initial Grid"
      [,1] [,2] [,3] [,4]
[1,] "S"   ""   ""   ""
[2,] ""   ""   ""   ""
[3,] ""   ""   ""   ""
[4,] ""   ""   ""   "G"

```

Starting Simulation

```

Step: 1
Current State : 1 1
Action Taken  : Down
Next State    : 2 1
Reward        : -1

```

```

Step: 2
Current State : 2 1
Action Taken  : Down
Next State    : 3 1
Reward        : -1

```

```

Step: 3
Current State : 3 1
Action Taken  : Down
Next State    : 4 1
Reward        : -1

```

```

Step: 4
Current State : 4 1
Action Taken  : Right
Next State    : 4 2
Reward        : -1

```

```

Step: 5
Current State : 4 2
Action Taken  : Right
Next State    : 4 3
Reward        : -1

```

Code 2:

R Program to Visualize an Agent's Path in a 4×4 Grid World

```
grid_size <- 4

x <- c(1,1,2,3,4,4,4)
y <- c(1,2,2,2,2,3,4)

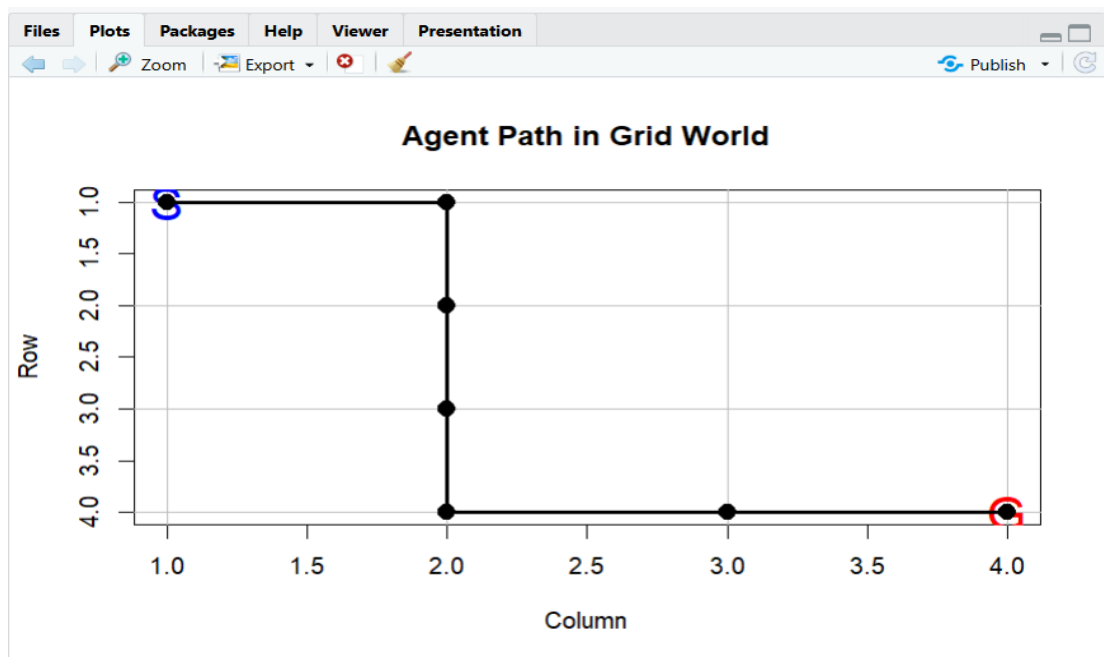
plot(
  1:grid_size,
  1:grid_size,
  type = "n",
  xlim = c(1,4),
  ylim = c(4,1),
  xlab = "Column",
  ylab = "Row",
  main = "Agent Path in Grid World"
)

abline(v = 1:4, col = "gray")
abline(h = 1:4, col = "gray")

text(1,1,"S",cex=2,col="blue")
text(4,4,"G",cex=2,col="red")

lines(y,x,lwd=3)
points(y,x,pch=19,cex=1.5)
```

OUTPUT:



Code 3:

#R Program to Visualize the Markov Decision Process (MDP) Workflow Using DiagrammeR

```
library(DiagrammeR)

grViz("
digraph MDP {

graph [
  layout = dot,
  rankdir = TB,
  bgcolor = '#FDFEFE',
  pad = 0.5,
  nodesep = 0.5,
  ranksep = 1.0,
  splines = ortho
]

node [
  shape = box,
  style = 'rounded, filled',
  fontname = 'Times New Roman',
  fontsize = 32,
  penwidth = 4,
  color = '#2C3E50',
  width = 3,
  height = 0.8
]

edge [
  color = '#5D6D7E',
  penwidth = 2,
  arrowsize = 0.9
]

A [label='Markov Decision Process', fillcolor='#1ABC9C', fontcolor='white']
B [label='State Space\n(S)', fillcolor='#5DADE2']
C [label='Action Space\n(A)', fillcolor='#F5B041']

D [label='Transition\nProbability', fillcolor='#58D68D']
E [label='Reward\nFunction', fillcolor='#EC7063']

F [label='Initialize\nPolicy', fillcolor='#AF7AC5']

G [label='Policy\nEvaluation', fillcolor='#85C1E9']
H [label='Policy\nImprovement', fillcolor='#F7DC6F']

I [label='Optimal\nPolicy', fillcolor='#2ECC71', fontcolor='white']
J [label='Decision\nMaking', fillcolor='#F1948A']

K [label='Performance\nEvaluation', fillcolor='#BB8FCE']

L [label='Final\nOutput', fillcolor='#95A5A6', fontcolor='white']

A -> B
A -> C
```

B -> D

C -> D

D -> E

E -> F

F -> G

G -> H

H -> G [label='No', color='red', fontcolor='red']

H -> I [label='Yes', color='darkgreen', fontcolor='darkgreen']

I -> J

J -> K

K -> L

}
")

OUTPUT :

